

Kunwoo Park

Ph.D. Student, Department of Electrical and Computer Engineering
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Research Interests

Integrated photonics; Disordered photonics; Topological photonics.

Education

Integrated M.S./Ph.D., Department of Electrical and Computer Engineering 2024.09 – present
Seoul National University Advisor: Sunkyu Yu (Expected: 2029.08)

B.S., Department of Electrical and Computer Engineering 2018.03 – 2024.08
Seoul National University

Military Service, Republic of Korea Army 2020.12 – 2022.06

Research Experience

- **Integrated Photonics.** Placeholder.
- **Disordered Photonics.** Placeholder.
- **Topological Photonics.** Placeholder.

Publications

Preprints

1. Jungyoon Kim†, Kyoho Kim†, **Kunwoo Park**, Namkyoo Park* & Sunkyu Yu*. Task-specific programming of chaos in neural circuits.
[arXiv:2605.19465](#)
2. Gitae Lee†, Seungmok Youn†, Ikbeom Lee†, **Kunwoo Park**, Duwhan Hwang, Xianji Piao*, Namkyoo Park* & Sunkyu Yu*. Non-Hermitian stealthy hyperuniformity.
[arXiv:2602.09458](#)
3. **Kunwoo Park**, Ikbeom Lee, Seungmok Youn, Gitae Lee, Namkyoo Park* & Sunkyu Yu*. Hypergraph modelling of wave scattering to speed-up material design.
[arXiv:2507.15329](#)

Journal Articles

1. Seungmok Youn, **Kunwoo Park**, Ikbeom Lee, Gitae Lee, Namkyoo Park* & Sunkyu Yu*. Phase-sensitive engineering of optical disordered materials using heterogeneous networks.
[Advanced Optical Materials](#) 14, e03499 (2026)
2. Kyuho Kim†, **Kunwoo Park**†, Hyunchul Park, Sunkyu Yu, Namkyoo Park* & Xianji Piao*. Programmable photonic unitary circuits for light computing.
[Nanophotonics](#) 14, 1429 (2025)

Conferences

1. Ikbeom Lee, **Kunwoo Park**, Seungmok Youn & Sunkyu Yu*. Inverse design of microstructures of non-Hermitian scattering systems.
SPIE Optics + Photonics 2025, 13578-59, San Diego, California, USA, 6 August 2025
2. **Kunwoo Park**, Ikbeom Lee, Seungmok Youn & Sunkyu Yu*. Modularization of light scattering for photonic computing applications.
SPIE Optics + Photonics 2025, 13581-25, San Diego, California, USA, 5 August 2025
3. Ikbeom Lee, **Kunwoo Park** & Sunkyu Yu*. Design of wave scattering from gain and loss disordered materials.
CLEO-PacificRim (CLEO-PR) 2024, Mo2J-4, Incheon, Republic of Korea, 5 August 2024
4. **Kunwoo Park**, Ikbeom Lee & Sunkyu Yu*. Inverse design method of hyperuniform materials using system factorization.

Patents

1. Nonlinear unitary optical device.

Sunkyu Yu, Namkyoo Park, Xianji Piao & **Kunwoo Park**

KR 10-2859166 (Application No. 10-2025-0005056; Provisional No. 10-2024-0179951)

Registration date: 2025.09.09; Filing date: 2025.01.13; Provisional filing date: 2024.12.05

US application: 19/048,370; Filing date: 2025.02.07

EU application: 25156862.2; Filing date: 2025.02.10

Grants and Fellowships

Graduate Research Encouragement Grant ()

2025 – present

Principal Investigator

Teaching

Application of Quantum Mechanics (Graduate), Teaching Assistant

Spring 2025

Seoul National University

Topics in Electro-physics (Graduate), Teaching Assistant

Spring 2025

Seoul National University

Technical Skills

- **Theory and modeling:** Placeholder.
- **Computation:** Placeholder.
- **Experimentation:** Placeholder.
- **Software:** Placeholder.

Professional Service

- Placeholder.

References

Available upon request.

{[†] Co-first authors. * Corresponding authors.}